

Session – 1**1. Name 5 IT Job roles and write one line about what each role does.**

- **AI Prompt Engineer:** Designs and refining text prompts to help artificial intelligence models produce accurate and useful answers.
- **Cloud Architect:** Designs and manages an organization's cloud computing strategy, including application architecture and deployment.
- **Cybersecurity Analyst:** Protects an organization's computer networks and systems by monitoring security breaches and investigating violations.
- **Data Scientist:** Analyzes complex data sets to help organizations make informed, data-driven business decisions.
- **DevOps Engineer:** Bridges the gap between software development and IT operations to speed up code deployment and system reliability.
- **Software Developer –** Develops, tests, and maintains software applications.
- **Web Developer –** Builds and maintains websites and web applications.
- **Database Administrator (DBA) –** Manages databases, including data storage, security, and backup.
- **Software Tester / QA Engineer –** Tests software to find bugs and ensure it works correctly.
- **Network Engineer –** Manages and maintains computer networks and network infrastructure.

2. List of 3 product companies and 3 service companies with one example of work each does.

No	Type	Companies	Example of Work
1	Product	Microsoft	Develop products such as Windows and Microsoft 365.
2	Product	Google	Develop products such as Google Search and Google Maps.
3	Product	Adobe	Develop software products such as Photoshop and Acrobat.
4	Service	TCS	Provides IT consulting, software development, and technology services to clients.
5	Service	Infosys	Provides IT consulting, application development, and digital services to clients
6	Service	Wipro	Provides IT services, consulting, and technology solutions to businesses.
7	Product	Odoo	Develops an integrated business management software suite for CRM, accounting, sales, inventory, HR, and other business operations.

3. Write the full forms and one-line use of GitHub, Slack, Jira, IDE, API.

No	Term	Full Form	One line use
1	GitHub	GitHub is a platform for hosting and collaborating on Git repositories.	Used to store, manage and collaborate on source code.
2	Slack	Slack is a team communication and collaboration platform.	Used for communication and sharing information within development teams.
3	Jira	Jira is a project and issue tracking tool developed by Atlassian.	Used to track tasks, bugs and project progress.
4	IDE	Integrated Development Environment	Used to write, run debug and manage program code.

5	API	Application Programming Interface	Allow different software applications to communicate with each other.
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4. Name any 5 programming languages and mention one popular use case for each.

No	Programming Languages	Use cases
1	Python	Used for Web Development, automation, data science and AI.
2	Java	Used for enterprise applications and Android development.
3	C++	Used for system software, game development, and performance intensive applications.
4	C#	Used for windows applications, web applications and game development with Unity.
5	JavaScript	Used for interactive websites and web applications.

5. What is difference between a developer and a tester?

- A developer designs and writes code to build software features and applications.
- A tester checks the software to find bugs and verify that it works according to requirements.

Session – 2

1. What are three main layers of software architecture? Write one line about each.

Well, there are total 5 layers in software architecture as below: and among them 3 main layers are Presentation Layer, Business Layer and Database Layer.

No	Layer	Descriptions
1	Presentation Layer	Handles the user interface and interaction b/w the user and application.
2	Application Layer	Handles the main program of architecture. It includes the code definition and most basic function of the developed application.
3	Business Layer	Also called domain layer where application's business logic operates.
4	Persistence Layer	Also called data access layer & act as a protective layer. It contains the code that's necessary to access the database layer.
5	Database Layer	Handles storing, retrieving and managing data from the database.

2. Define client and server in your own words with one real-life example.

Client	A client is a device or application that req data from a server.
Server	Server is a computer system who receive requests from clients and provides required data.
Real life example	Ex – 1 When you prompt a ChatGPT using your phone or laptop your device act as client and ChatGPT 's model act as a server for providing data and information. Ex – 2 When you open YouTube on your phone, your phone acts as the client and YouTube's servers provide the requested videos and data.

3. Write 2 differences between desktop apps, web apps, and mobile apps.

Desktop Apps	Web Apps	Mobile Apps
Install directly on computer	Access through a web browser	Designed mainly for smart phones and tablets.
Usually designed for windows, macOS, Linux OS	Usually does not require an installation on user's device	Usually installed through Google play stores or Apple app stores or other 3 RD party.

4. Name any 3 database and mention whether each is SQL or NOSQL.

NO	Database	Type
1	PostgreSQL	SQL
2	MySQL	SQL
3	MongoDB	NoSQL
4	CockroachDB	SQL

Session – 3**1. Write the full form and one line use of: HTTP, HTTPS, FTP, SMTP.**

Term	Full form	One line use
HTTP	HyperText Transfer Protocol	Used to transfer web pages and other data b/w a client and web server.
HTTPS	HyperText Transfer Protocol Secure	Used to securely transfer data between a client and web server using encryption.
FTP	File Transfer Protocol	Used to transfer files between computers over a network.
SMTP	Simple Mail transfer protocol	Used to send emails from one mail server to another or from a client to a mail server.

2. What is the difference b/w IPV4 and IPV6? Answer in 2 lines.

- IPV4 uses a 32-bit address and supports about 4.3 billion unique addresses, written like **192.168.1.1**
- IPv6 uses a 128 bit address, providing a much larger number of addresses, written like **2001:db8::1**

3. Run nslookup google.com on your system and write down the IP address you get.

```
C:\Users\Mr. Kevin>nslookup google.com
Server: UnKnown
Address: 10.224.185.80

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4002:831::200e
           172.217.26.110

C:\Users\Mr. Kevin>|
```

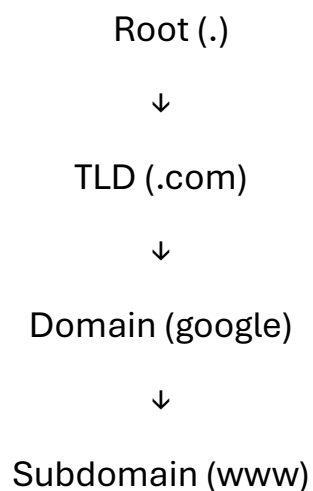
4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Code	Meaning	Explanation
200	OK	The request was successful.
301	Moved Permanently	The requested resource has been permanently moved to a new URL.
404	Not Found	The requested resource or webpage could not be found.
500	Internal Server Error	The server encountered an unexpected problem while processing the request.
403	Forbidden	The server understood the request but refuses to allow access.

5. Explain DNS hierarchy.

DNS (Domain name system) uses a hierarchy to translate domain names into IP addresses.

For www.google.com



- Root (.) – The top level of the dns hierarchy.
- TLD(.com) – The top-level domain.
- Domain (google) -- The registered domain name.
- Subdomain (www) – A sub domain used to identify a particular service or host under google.com.

Session – 4**1. List all 6 phases of SDLC in order with one line about each.**

- Requirement Gathering: - Collect and document the customer's needs and project requirements.
- Analysis: - Analysis the requirements to determine the project scope, feasibility, and resource needed.
- Designing: - Create the system architecture, database and UI Design, and technical specifications.
- Implementation: - Developers write the code and build the software according to the design.
- Testing: - Test the software to find and fix bugs and ensure it meet the requirements.
- Maintenance: - Maintain and update the software after deployment to fix issues and improvements.

2. Write 3 differences between Agile and Waterfall models.

Agile Models	Waterfall models
Development is iterative and flexible.	Development follows a fixed sequential process.
Requirements can change during development.	Requirements are usually defined before development and change are difficult.
Testing is performed continuously throughout development.	Testing is generally performed after the development phase.

3. Name the 3 main Scrum roles and write one line about the responsibility of each.

Three main scrum roles	
Product Owner	Define product requirements, manage the product backlog, and prioritize work.
Scrum Master	Helps the team to follow Scrum practices and remove obstacles.
Developers/ Development Team	Design, develop, test, and deliver the product increment.

4. What is a Sprint in Scrum? Mention its typical duration.

- A sprint is a fixed period during which the scrum team works on one selected task and delivers a usable product increment.
- Typical duration: 2 weeks (can range from 1-4 weeks).

5. What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

- **Backlog** – List of all pending tasks/user stories that may be worked on later.
- **To Do** – Tasks selected for work but not started yet.
- **In Progress** – Tasks currently being worked on.
- **Done** – Tasks that have been completed and meet the required criteria.

Session – 5

1. Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

- `git init` → for initialize a repo
- `git add` → Stage all files
- `git commit -m "Message"` → Commit with a msg
- `git push origin main` → push to remote
- `git pull origin main` → pull latest changes

2. What is the difference between git fetch and git pull? Answer in 2 lines.

- Git fetch downloads the latest changes from the remote repo but does not modify your working files.
- Git pull downloads the latest changes and merges them into your current branch.

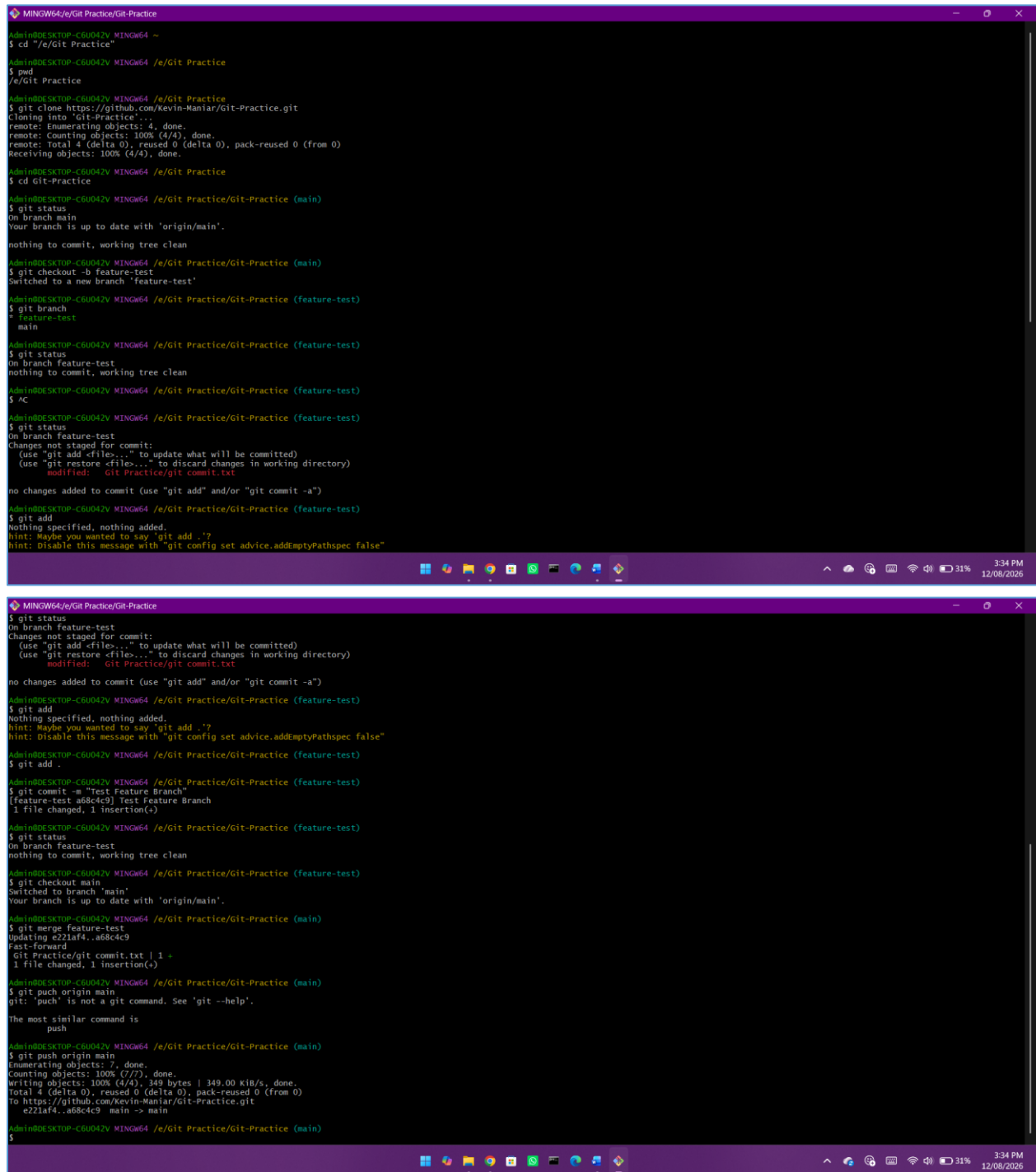
3. Create a new GitHub repository, push any small text file from your local system, and share the repo link.

<https://github.com/Kevin-Maniar/Git-Practice>

4. What is the difference between a Fork and a Pull Request? Answer in 2 lines.

- A fork creates your own copy of someone else's GitHub repo under your GitHub Account.
- A Pull req is a request to merge your changes from your branch/fork into another repo's branch.

5. Create a new branch named feature-test, make one commit on it, and merge it back to the main branch. Write the commands you used.



```

MINGW64/e/Git Practice/Git-Practice
Admin@DESKTOP-CE0U42V MINGW64 ~
$ cd "/e/Git Practice"
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice
$ pwd
/e/Git Practice
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice
$ git clone https://github.com/kevin-Maniar/Git-Practice.git
Cloning into 'Git-Practice'...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice
$ cd Git-Practice
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (main)
$ git checkout -b feature-test
Switched to a new branch 'feature-test'
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git branch
* feature-test
  main
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git status
On branch feature-test
nothing to commit, working tree clean
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ AC
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git status
On branch feature-test
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   Git Practice/git commit.txt
no changes added to commit (use "git add" and/or "git commit -a")
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git add
Nothing specified, nothing added.
hint: Maybe you wanted to say "git add ."?
hint: Disable this message with "git config set advice.addEmptyPathsSpec false"
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git add .
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git commit -m "Test Feature Branch"
[feature-test a68c4c9] Test Feature Branch
1 file changed, 1 insertion(+)
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git status
On branch feature-test
nothing to commit, working tree clean
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (feature-test)
$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (main)
$ git merge feature-test
Updating e221af4..a68c4c9
Fast-forward
 Git Practice/git commit.txt | 1 +
1 file changed, 1 insertion(+)
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (main)
$ git push origin main
git: 'push' is not a git command. See 'git --help'.

The most similar command is
  push
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (main)
$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (4/4), done.
Writing objects: 100% (4/4), 349 bytes | 349.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/kevin-Maniar/Git-Practice.git
e221af4..a68c4c9  main -> main
Admin@DESKTOP-CE0U42V MINGW64 /e/Git Practice/Git-Practice (main)
$
  
```